

chapter-9

1. (i) Let, $A = \begin{bmatrix} 3 & 1 \\ 5 & 7 \end{bmatrix}$

The characteristic polynomial of A is,

$$\begin{aligned} \lambda I - A &= \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 5 & 7 \end{bmatrix} \\ &= \begin{bmatrix} \lambda-3 & -1 \\ -5 & \lambda-7 \end{bmatrix} \end{aligned}$$

Now, the characteristic equation is,

$$|\lambda I - A| = 0$$

$$\Rightarrow \begin{vmatrix} \lambda-3 & -1 \\ -5 & \lambda-7 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} \lambda-3 & -1 & 0 \\ \lambda-3 & -1 & 0 \\ \lambda-3 & -1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 - 3\lambda - 7\lambda + 21 - 5 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda + 16 = 0$$

$$\Rightarrow (\lambda-8)(\lambda-2) = 0$$

$$\therefore \lambda = 8, 2$$

Thus the eigenvalues are 8, 2

QD. Let, $A = \begin{bmatrix} -3 & 2 & 2 \\ -6 & 5 & 2 \\ -7 & 4 & 4 \end{bmatrix}$

The characteristic polynomial of A is, (1.1)

$$\begin{aligned} \lambda I - A &= \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} -3 & 2 & 2 \\ -6 & 5 & 2 \\ -7 & 4 & 4 \end{bmatrix} \\ &= \begin{bmatrix} \lambda+3 & -2 & -2 \\ 6 & \lambda-5 & -2 \\ 7 & -4 & \lambda-4 \end{bmatrix} \end{aligned}$$

The characteristic equation is,

$$|\lambda I - A| = 0$$

$$\therefore \begin{vmatrix} \lambda+3 & -2 & -2 \\ 6 & \lambda-5 & -2 \\ 7 & -4 & \lambda-4 \end{vmatrix} = 0$$

$$\Rightarrow (\lambda+3)(\lambda^2-4\lambda-5\lambda+20) + 2(6\lambda-24) - 2(-24-7\lambda+35) = 0$$

$$\Rightarrow (\lambda+3)(\lambda^2-9\lambda+10) + 12\lambda-20-22+14\lambda = 0$$

$$\Rightarrow \lambda^3-9\lambda^2+10\lambda+3\lambda^2-27\lambda+45+26\lambda-42 = 0$$

$$\Rightarrow \lambda^3-6\lambda^2+13\lambda-6 = 0$$

$$\Rightarrow \lambda^3-\lambda^2-5\lambda^2+5\lambda+6\lambda-6 = 0$$

$$\Rightarrow \lambda^2(\lambda-1)-5\lambda(\lambda-1)+6(\lambda-1) = 0$$

$$\Rightarrow (\lambda-1)(\lambda^2-5\lambda+6) = 0$$

$$\Rightarrow (\lambda-1)(\lambda-3)(\lambda-2) = 0 \quad \therefore \lambda = 1, 3, 2$$

Thus the eigenvalues are 1, 2, 3.

2. Let, $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$

The characteristic polynomial is,

$$\begin{aligned} \lambda I - A &= \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \\ &= \begin{bmatrix} \lambda - 1 & -2 \\ -3 & \lambda - 2 \end{bmatrix} \end{aligned}$$

The characteristic equation is

$$|\lambda I - A| = 0$$

$$\therefore \begin{vmatrix} \lambda - 1 & -2 \\ -3 & \lambda - 2 \end{vmatrix} = 0$$

$$\lambda^2 - \lambda - 2\lambda + 2 - 6 = 0$$

$$\Rightarrow \lambda^2 - 3\lambda - 4 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + \lambda - 4 = 0$$

$$\therefore (\lambda - 4)(\lambda + 1) = 0$$

$$\therefore \lambda = 4, -1$$

Thus the eigenvalues are $\begin{bmatrix} 4 & -1 \\ -1 & 4 \end{bmatrix}$

$$\text{Let, } v = \begin{bmatrix} x \\ y \end{bmatrix}$$

Then by definition we get,

$$Av = \lambda v$$

$$\Rightarrow (\lambda I - A)v = 0 \quad \text{--- (1)}$$

putting $\lambda = -1$ in (1) we get,

$$\begin{bmatrix} -2 & -2 \\ -3 & -3 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\therefore -2x - 2y = 0$$

$$-3x - 3y = 0$$

Reduce this system to echelon form,

$$-2x - 2y = 0$$

$$0 = 0 \quad | \quad -3L_1 + 2L_2$$

Here y is a free variable.

$$\text{Let, } y = -1 \therefore x = 1$$

Hence, eigenvector corresponding to, $\lambda = -1$ is $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

putting $\lambda = 4$ in ① we get,

$$\begin{bmatrix} 3 & -2 \\ -3 & 2 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow 3x - 2y = 0$$

$$-3x + 2y = 0$$

Reduce this system to echelon form,

$$3x - 2y = 0$$

$$0 = 0 \quad | \quad L_1 + L_2$$

Here y is a free variable.

$$\text{let, } y = 3 \quad \therefore x = 2$$

Hence, the corresponding to $\lambda = 4$, eigenvector is $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

3. (iii) Let, $A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$

The characteristic polynomial of A is

$$\begin{aligned} \lambda I - A &= \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} \lambda - 3 & -2 & -4 \\ -2 & \lambda & -2 \\ -4 & -2 & \lambda - 3 \end{bmatrix} \end{aligned}$$

The characteristic equation is,

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda-3 & -2 & -4 \\ -2 & \lambda & -2 \\ -4 & -2 & \lambda-3 \end{vmatrix} = 0$$

$$\Rightarrow (\lambda-3)(\lambda^2-3\lambda-4) + 2(-2\lambda+6-8) - 4(4+4\lambda) = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 - 4\lambda - 3\lambda^2 + 9\lambda + 12 - 4\lambda - 4 - 16 - 16\lambda = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 - 15\lambda - 8 = 0$$

$$\Rightarrow \lambda^3 - 8\lambda^2 + 2\lambda^2 - 16\lambda + \lambda - 8 = 0$$

$$\Rightarrow \lambda^2(\lambda-8) + 2\lambda(\lambda-8) + 1(\lambda-8) = 0$$

$$\Rightarrow (\lambda-8)(\lambda^2+2\lambda+1) = 0$$

$$\Rightarrow (\lambda-8)(\lambda+1)^2 = 0$$

$$\therefore \lambda = 8, -1, -1$$

Thus the eigenvalues are 8, -1, -1

$$\text{Let } v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

By the definition we get,

$$\lambda v = Av$$

$$\Rightarrow (\lambda I - A)v = 0$$

$$\therefore \begin{bmatrix} \lambda-3 & -2 & -4 \\ -2 & \lambda & -2 \\ -4 & -2 & \lambda-3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \quad \text{--- (1)}$$

putting $\lambda = -1$ in ① we get,

$$\begin{bmatrix} -4 & -2 & -4 \\ -2 & -1 & -2 \\ -4 & -2 & -4 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\therefore \begin{aligned} -4x - 2y - 4z &= 0 \\ -2x - y - 2z &= 0 \\ -4x - 2y - 4z &= 0 \end{aligned}$$

Reduce this system to echelon form,

$$\begin{aligned} -4x + 2y + 4z &= 0 \\ 0 &= 0 \\ 0 &= 0 \end{aligned} \quad \left| \begin{array}{l} -L_1 + 2L_2 \\ -L_1 + L_3 \end{array} \right.$$

Here y and z are free variables.

Let (i) $y = 2, z = -1 \therefore x = 0$

Hence the ~~corresponding~~ eigenvector corresponding to $\lambda = -1$

$$\text{is } v_1 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$$

(ii) $y = 0, z = -1 \therefore x = 1$

Hence the eigenvector corresponding to

$$\lambda = -4 \text{ is } v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

putting, $\lambda = 8$ in ① we get,

$$\begin{bmatrix} 5 & -2 & -4 \\ -2 & 8 & -2 \\ -4 & -2 & 5 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\begin{bmatrix} 5 & -2 & -4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = A - \lambda I$$

$$\therefore 5x - 2y - 4z = 0$$

$$-2x + 8y - 2z = 0$$

$$-4x - 2y + 5z = 0$$

Reduce this system to echelon form,

$$\begin{array}{l} 5x - 2y - 4z = 0 \\ 36y - 18z = 0 \\ -18y + 9z = 0 \end{array} \left| \begin{array}{l} 2L_1 + 5L_2 \\ 4L_1 + 5L_3 \end{array} \right.$$

OR,

$$5x - 2y - 4z = 0$$

$$36y - 18z = 0$$

$$0 = 0 \mid L_2 + 2L_3$$

Here z is a free variable.

$$\text{let, } z = 2, \therefore y = 1, x = 2$$

Hence the eigenvector corresponding to

$$\lambda = 8 \text{ is } \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

4. (a)

$$A = \begin{bmatrix} -1 & -5 & 7 \\ -3 & -5 & 9 \\ -3 & -9 & 13 \end{bmatrix}$$

$$0 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \times \begin{bmatrix} \lambda - 1 & -5 & 7 \\ -3 & -5 & 9 \\ -3 & -9 & 13 \end{bmatrix}$$

The characteristic polynomial of A is,

$$\lambda I - A = \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} -1 & -5 & 7 \\ -3 & -5 & 9 \\ -3 & -9 & 13 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda+1 & 5 & -7 \\ 3 & \lambda+5 & -9 \\ 3 & 9 & \lambda-13 \end{bmatrix}$$

The characteristic equation is,

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda+1 & 5 & -7 \\ 3 & \lambda+5 & -9 \\ 3 & 9 & \lambda-13 \end{vmatrix} = 0$$

$$\Rightarrow (\lambda+1)(\lambda^2 - 13\lambda + 5\lambda - 65 + 81) - 5(3\lambda - 39 + 27) - 7(27 - 3\lambda - 15) = 0$$

$$\Rightarrow (\lambda+1)(\lambda^2 - 8\lambda + 16) - 5(3\lambda - 12) - 7(12 - 3\lambda) = 0$$

$$\Rightarrow \lambda^3 - 8\lambda^2 + 16\lambda + \lambda^2 - 8\lambda + 16 - 15\lambda + 60 - 84 + 21\lambda = 0$$

$$\Rightarrow \lambda^3 - 7\lambda^2 + 14\lambda - 8 = 0$$

$$\Rightarrow \lambda^3 - \lambda^2 - 6\lambda^2 + 6\lambda + 8\lambda - 8 = 0$$

$$\Rightarrow \lambda^2(\lambda - 1) - 6\lambda(\lambda - 1) + 8(\lambda - 1)$$

$$\Rightarrow (\lambda-1)(\lambda^2-6\lambda+8) = 0$$

$$\Rightarrow (\lambda-1)(\lambda^2-4\lambda-2\lambda+8) = 0$$

$$\Rightarrow (\lambda-1)(\lambda-4)(\lambda-2) = 0$$

$$\therefore \lambda = 1, 2, 4$$

Again, from ① we get,

$$\lambda^3 - 7\lambda^2 + 14\lambda - 8 = 0$$

By Cayley-Hamilton theorem, we get

$$A^3 - 7A^2 + 14A - 8I = 0$$

$$\Rightarrow 8I = A^3 - 7A^2 + 14A$$

Multiply both side with A^{-1} .

$$\Rightarrow 8IA^{-1} = A^3 \cdot A^{-1} - 7A^2 \cdot A^{-1} + 14A \cdot A^{-1}$$

$$\Rightarrow 8A^{-1} = A^2 - 7A + 14I$$

$$\therefore A^{-1} = \frac{1}{8} (A^2 - 7A + 14I)$$

[proved]

$$4. (b) \quad B = \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix}$$

The characteristic polynomial of B is,

$$\begin{aligned} \lambda I - B &= \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix} \\ &= \begin{bmatrix} \lambda & c & -b \\ -c & \lambda & a \\ b & -a & \lambda \end{bmatrix} \end{aligned}$$

The characteristic equation is,

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda & c & -b \\ -c & \lambda & a \\ b & -a & \lambda \end{vmatrix} = 0$$

$$\lambda(\lambda^2 + a^2) - c(-c\lambda - ab) - b(ac - b\lambda) = 0$$

$$\Rightarrow \lambda^3 + a^2\lambda + c^2\lambda + abc - abc + b^2\lambda = 0$$

$$\Rightarrow \lambda^3 + \lambda(a^2 + b^2 + c^2) = 0$$

By Cayley-Hamilton theorem,

$$B^3 + B(a^2 + b^2 + c^2) = 0$$

76. (a) let,

$$A = \begin{bmatrix} 2 & -3 & 4 & 1 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & -2 & 3 \end{bmatrix}$$

$$0 = \begin{vmatrix} \lambda - 2 & 3 & -4 & -1 \\ 0 & \lambda - 3 & -1 & -2 \\ 0 & 0 & \lambda - 1 & -5 \\ 0 & 0 & 2 & \lambda - 3 \end{vmatrix}$$

The characteristic polynomial of A is,

$$\lambda I - A = \lambda \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -3 & 4 & 1 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda - 2 & 3 & -4 & -1 \\ 0 & \lambda - 3 & -1 & -2 \\ 0 & 0 & \lambda - 1 & -5 \\ 0 & 0 & 2 & \lambda - 3 \end{bmatrix}$$

The characteristic equation is,

$$|\lambda I - A| = 0$$

$$\begin{vmatrix} \lambda - 2 & 3 & -4 & -1 \\ 0 & \lambda - 3 & -1 & -2 \\ 0 & 0 & \lambda - 1 & -5 \\ 0 & 0 & 2 & \lambda - 3 \end{vmatrix} = 0$$

$$\Rightarrow (\lambda - 2) \begin{vmatrix} \lambda - 3 & -1 & -2 \\ 0 & \lambda - 1 & -5 \\ 0 & 2 & \lambda - 3 \end{vmatrix} - 3 \begin{vmatrix} 0 & -1 & -2 \\ 0 & \lambda - 1 & -5 \\ 0 & 2 & \lambda - 3 \end{vmatrix} - 4 \begin{vmatrix} 0 & \lambda - 3 & -2 \\ 0 & 0 & -5 \\ 0 & 0 & \lambda - 3 \end{vmatrix}$$

$$+ \begin{vmatrix} 0 & \lambda-3 & -1 \\ 0 & 0 & \lambda-1 \\ 0 & 0 & 2 \end{vmatrix} = 0$$

$$\Rightarrow (\lambda-2) \{ (\lambda-3) [(\lambda-1)(\lambda-3) + 10] + 0 - 0 \} + -3 \{ 0 + 1 \cdot 0 \} - 0 = 0$$

$$\Rightarrow (\lambda-2) \{ (\lambda-3) (\lambda^2 - 3\lambda - \lambda + 3 + 10) \} = 0$$

$$\Rightarrow (\lambda-2) \{ (\lambda-3) (\lambda^2 - 4\lambda + 13) \} = 0$$

$$\Rightarrow (\lambda-2) (\lambda-3) (\lambda^2 - 4\lambda + 13) = 0$$

$$\begin{bmatrix} 1-\lambda & p & c & s-\lambda \\ s & 1-\lambda & c-\lambda & 0 \\ 2 & 1-\lambda & 0 & 0 \\ c-\lambda & s & 0 & 0 \end{bmatrix} =$$

$$0 = |A - \lambda I|$$

$$\begin{vmatrix} 1-\lambda & p & c & s-\lambda \\ s & 1-\lambda & c-\lambda & 0 \\ 2 & 1-\lambda & 0 & 0 \\ c-\lambda & s & 0 & 0 \end{vmatrix}$$